1,580 AI Authors

Results of the December 2024 survey of AI researchers (Expert Survey on Progress in AI)

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Survey rounds and respondents

The same questions have been put to AI researchers in four rounds.

1,580

AI researchers

who published in 2023
at NeurIPS, ICML, ICLR,
AAAI, JMLR or IJCAI



1. Asked in December 2024.

2. Largely the same questions as in 2023.

3. A 10% response rate.

4. To the authors' knowledge, the longest-running series of such surveys.

Grace et al., Advanced AI according to 1,580 researchers: uncertain, unsafe, and sooner than we thought, AI Impacts, September 2026, aiimpacts.org/wp-content/uploads/2026/09/ESPAI2024.pdf.

Outline

The results are presented in four sections.

1. Automation of labor
2. Forecast of milestones
3. Expected outcomes and risk
4. Concerns and safety research

1

Section 1

Automation of labor

Forecast dates for human-level machine intelligence and for the full automation of labor

Definition of human-level machine intelligence

The definition every respondent was given.

> *"When unaided machines can accomplish every task better and more cheaply than human workers."*

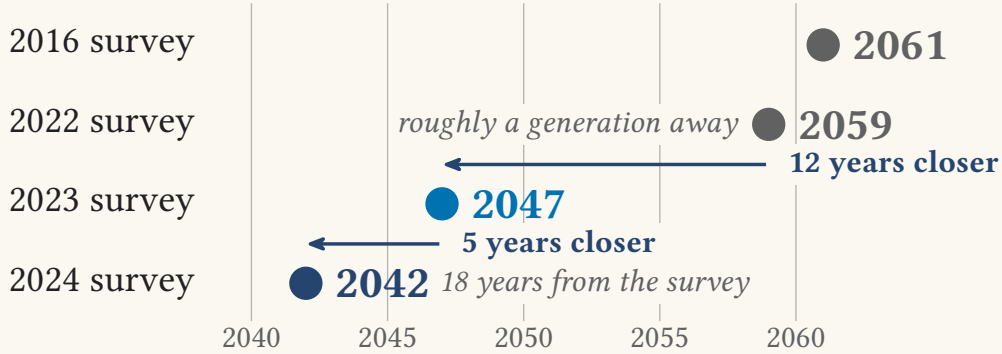
Think feasibility, not adoption.

1. They call it high-level machine intelligence, HLMI.

2. Call it AGI if you prefer.

Forecast date of human-level machine intelligence

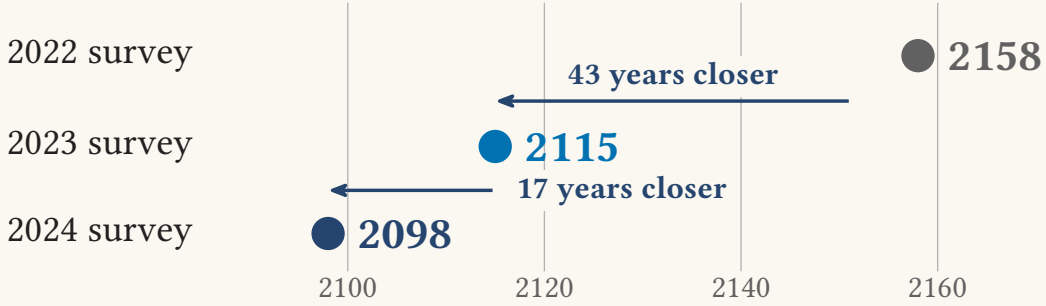
Year at which the aggregate forecast reaches 50%, in each survey round.



The 2022 date was reported as 2060 in the 2023 paper; the 2024 report recomputes it as 2059.

Forecast date of full automation of labor

FAOL, full automation of labor: all occupations fully automatable.



2

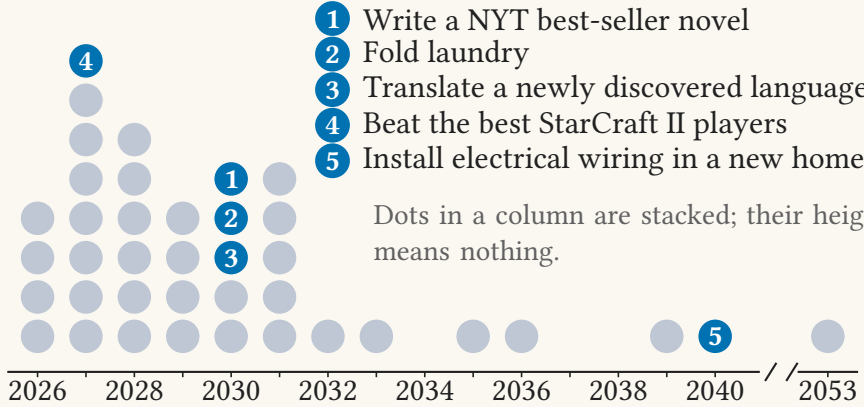
Section 2

Forecast of milestones

Fifty percent forecast dates for 39 specific tasks

Forecast dates for 39 tasks

Each dot is a task, placed at the year its forecast reaches 50%.



Tasks with the earliest forecast dates

The four tasks with a 50% date in 2026, the earliest of the 39.

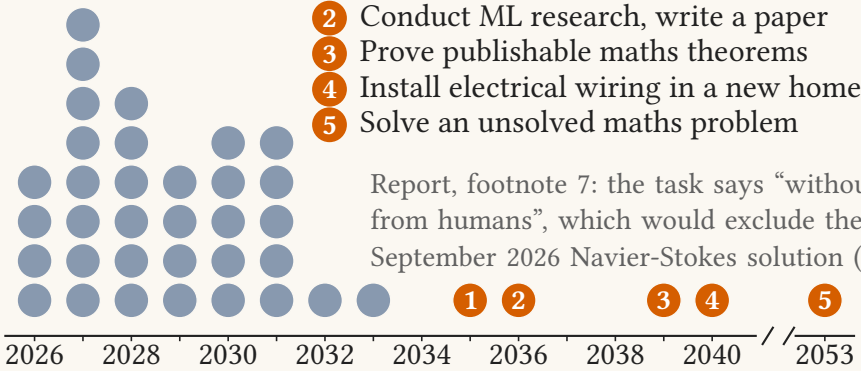


1. Write a high-school history essay — **2026**
2. Write Python code (e.g. quicksort) — **2026**
3. Play new Angry Birds levels (superhuman) — **2026**
4. Win World Series of Poker — **2026**

The report: for tasks with very short timelines "it is plausible that the task is already feasible".

34 of 39 tasks reach 50% probability by 2034

Ten years from the survey is 2034. Five tasks come later.



1. Discover physics from simulation — **2035**
2. Conduct ML research, write a paper — **2036**
3. Prove publishable maths theorems — **2039**
4. Install electrical wiring in a new home — **2040**
5. Solve an unsolved maths problem — **2053**

Report, footnote 7: the task says “without more input from humans”, which would exclude the proposed September 2026 Navier-Stokes solution (OpenAI, 2026).

3

Section 3

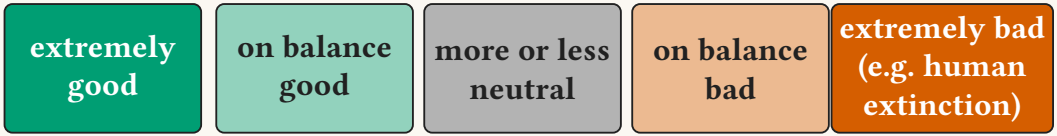
Expected outcomes and risk

Probabilities assigned to good and bad outcomes, and to extinction

Expected long-run impact of human-level AI

The exact question, asked of all respondents (n = 1,538).

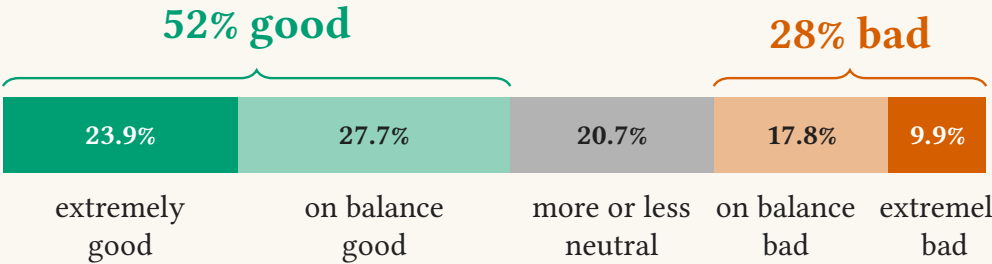
> *"Assume for the purpose of this question that HLMI will at some point exist. How positive or negative do you expect the overall impact of this to be on humanity, in the long run?"*



Respondents split 100% across the five kinds of impact.

Average probability of each kind of impact

The five kinds of impact, averaged over all respondents.



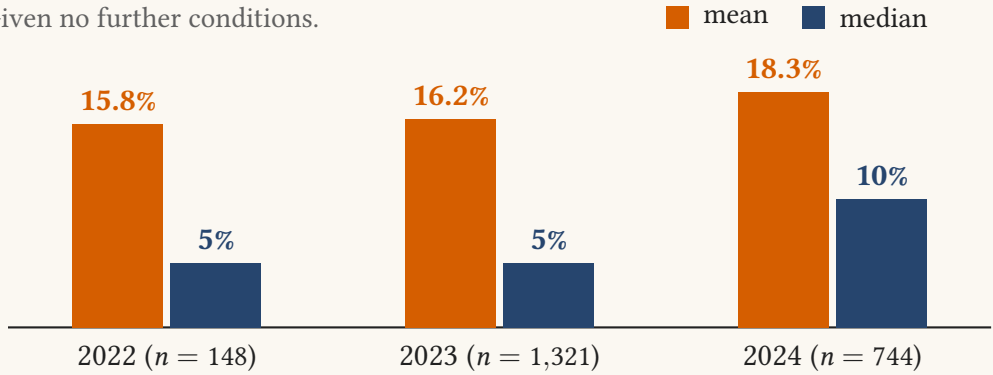
1. The median respondent puts 5% on extremely bad.

2. 788 of 1,538 respondents put at least 5% on both extremes.

Probability of extinction or disempowerment

Human extinction or similarly permanent and severe disempowerment.

Given no further conditions.



For the first time in these surveys, the median is 10%.

4

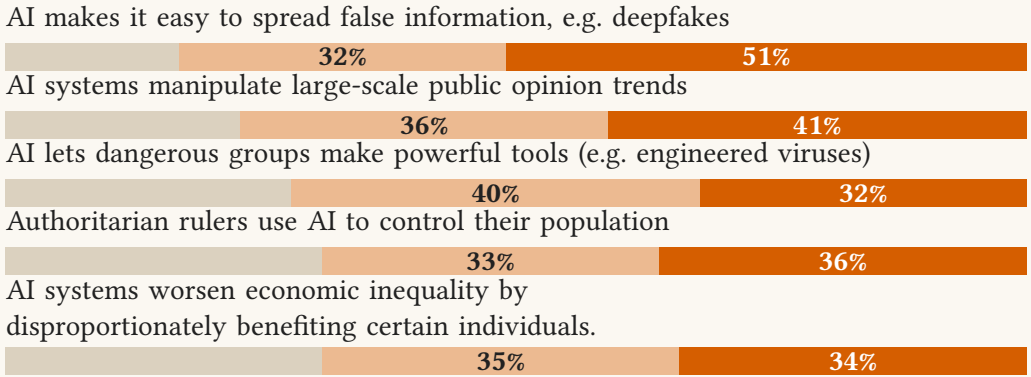
Section 4

Concerns and safety research

Scenarios of concern, expected traits of AI systems and the priority of safety research

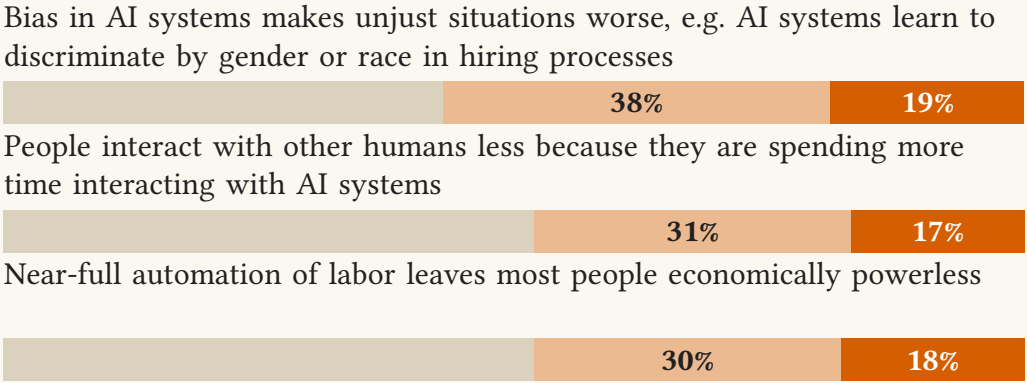
Concern about eleven scenarios, 1 of 3

Concern over the next thirty years. **deepfakes: 83% substantial or extreme**



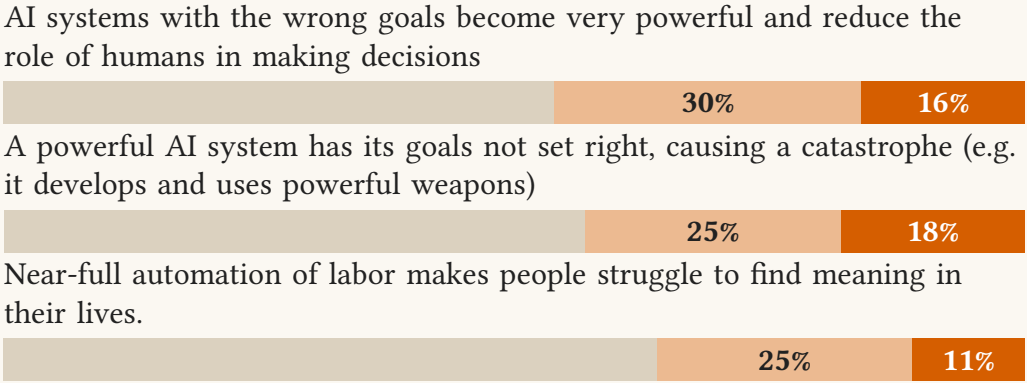
Concern about eleven scenarios, 2 of 3

Concern over the next thirty years, continued.



Concern about eleven scenarios, 3 of 3

Least concern of the eleven: substantial or extreme for over a third.



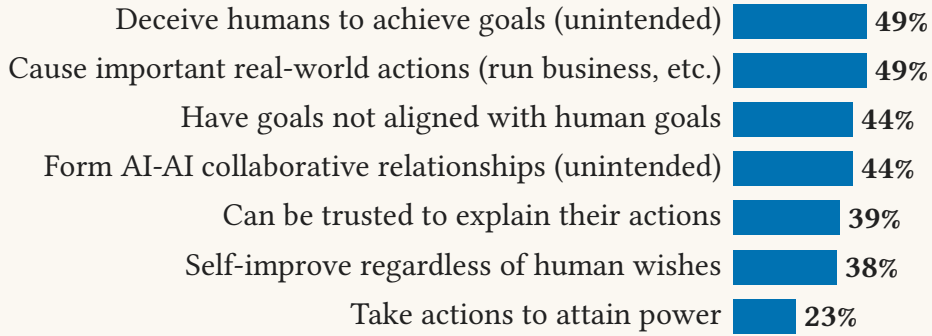
Traits expected of AI systems in 2044, 1 of 2

Share who rate each trait likely or very likely for some state-of-the-art AI.



Traits expected of AI systems in 2044, 2 of 2

The same question: likely or very likely in 2044.

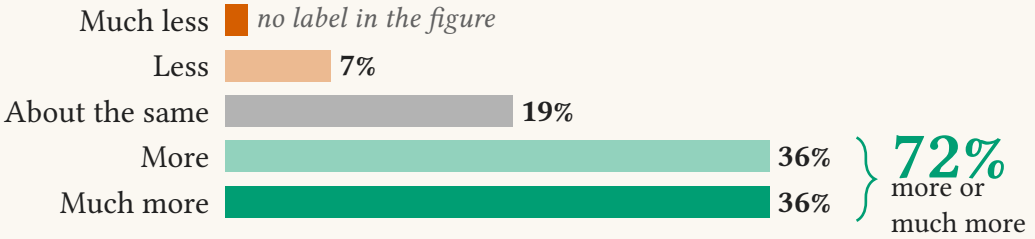


Up since 2023 for almost every trait.

Priority of AI safety research

Asked of half of the respondents (n = 748).

"How much should society prioritize AI safety research, relative to how much it is currently prioritized?"



The median respondent answers "more". Very similar to the 2022 and 2023 surveys; around 20 percentage points above the 2016 survey.

Quotations and definitions (1)

- **HLMI definition:** “unaided machines can accomplish every task better and more cheaply than human workers.” (first letter capitalised on the slide) and “Think feasibility, not adoption.” Report p. 5; the question is on p. 12.
- **FAOL definition:** “when all occupations are fully automatable.” Report p. 5.
- **Outcomes question:** “Assume for the purpose of this question that HLMI will at some point exist. How positive or negative do you expect the overall impact of this to be on humanity, in the long run?” Report p. 19, asked of all respondents (n=1538).
- **Five kinds of impact:** “Extremely good (e.g. rapid growth in human flourishing) … On balance good … More or less neutral … On balance bad … Extremely bad (e.g. human extinction)” Report p. 21.

Quotations and definitions (2)

- **Very short timelines:** “it is plausible that the task is already feasible” Report p. 10.
- **Footnote 7:** “without more input from humans” and “the proposed September 2026 Navier-Stokes solution” Report p. 9, footnote 7.
- **Two scenarios:** “AI makes it easy to spread false information, e.g. deepfakes” and “AI systems manipulate large-scale public opinion trends” Report p. 19; the 83% is in the abstract, p. 1; Figure 12, p. 21.
- **AI safety research:** the question of the slide is quoted from p. 28 (asked of half of the participants, n=748). The definition given to the respondents: “any AI-related research that, rather than being primarily aimed at improving the capabilities of AI systems, is instead primarily aimed at minimizing potential risks of AI systems” Report p. 5.

Where each date comes from

- **HLMI, 50% date:** 2061 (2016), 2059 (2022), 2047 (2023), 2042 (2024). Report p. 1, footnote 1: 2022 was reported as 2060 in the 2023 paper. The drops of twelve and five years are on p. 12.
- **FAOL, 50% date:** 2158 (2022), 2115 (2023), 2098 (2024). Report p. 12, footnote 8: 2023 was reported as 2116 in the 2023 paper. The drops of 43 and 17 years are on p. 12.
- **The 39 tasks:** the 50% year of each is in Table 7, report p. 45; 34 of 39 by 2034 and the five later ones are on p. 9.

Where each chart comes from

- **Section 2, the dots:** one dot per row of Table 7, report p. 45, at its printed 50% year.
- **Section 3:** Table 3, p. 22, and the text of p. 21 (the 52%, the 28% and the 788 of 1,538 respondents); Figure 17, p. 26 (means and medians).
- **Section 4:** the percentages are printed inside Figures 10, 12 and 21 of the report (pp. 20, 21, 30); the text repeats the 83% and the 72% (pp. 1, 28, 33).
- **Every chart is redrawn** from these numbers; none of the report's figures is reproduced.

The two papers

- Grace, K., Simonelli, J., Weinstein-Raun, B., Krueger, D., Young, N., Duffy, T., Adamczewski, T., Reeve, J. and Beck, A. (2026). *Advanced AI according to 1,580 researchers: uncertain, unsafe, and sooner than we thought.* AI Impacts, September 2026, 48 pages. The 2024 round of the Expert Survey on Progress in AI. A report on the AI Impacts site; the PDF states no licence.

- Grace, K., Sandkühler, J. F., Stewart, H., Weinstein-Raun, B., Thomas, S., Stein-Perlman, Z., Salvatier, J., Brauner, J. and Korzekwa, R. C. (2025). *Thousands of AI Authors on the Future of AI.* Journal of Artificial Intelligence Research, 84. doi:10.1613/jair.1.19087. The 2023 round, with 2,778 respondents; the earlier deck was built on it.